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Introduction to Llama 2: A High Quality Open Source Large Language Model

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Course Syllabus

Welcome

Introduction to the Course

Welcome to *Llama for Python Programmers*, a course designed for software developers eager to explore and harness the capabilities of Llama 2, an open access Large Language Model (LLM). Whether you're a seasoned Python programmer or someone who is increasing their proficiency, this course is structured to elevate your skills by introducing you to the integration of Llama 2 within your Python projects.

Success in this course comes from a combination of enthusiasm, and active participation. We encourage you to engage with the course materials, utilize the Coursera online lab space, and apply what you learn in practical exercises designed to reinforce your understanding and skills.

Introduction to Instructor

Christopher Brooks is an Associate Professor in the School of Information at the University of Michigan. His research focus is on the design of tools to better the teaching and learning experience in higher education, with a particular interest in understanding how learning analytics can be applied to human computer interaction through educational data mining, machine learning, and information visualization.

Learning Objective

By the end of this course you will be able to integrate Llama 2 into a Python-based software application using the llama.cpp and the related Python bindings.

Tools & Technology

Jupyter Notebooks (VS Code on Coursera Labs)

In this course, [the tool VS Code](#) is used to provide learners a programming environment to work in, directly on the Coursera platform through Coursera Labs. This environment utilizes Jupyter Notebooks to combine code and text, presented in an online programming environment, in a format that is compatible with both VSCode and the Jupyter Notebook tool.

This environment is intended to provide a ready-made workspace for learners to interact with course materials and the Llama models without the need for local setup. Please note that due to the performance requirements of large-language models, executing code that uses the Llama model may take some time to complete. These notebooks are available for learners to run locally, however support can only be provided for learners that utilize the Coursera Labs environment.

Course Schedule & Grading Policy

Module 1: Introduction to Llama 2: a High Quality Open Source LLM introduces you to Llama 2, highlighting its architecture, training method, and capabilities as a high-quality open-source LLM. This foundational segment prepares you for hands-on learning in the following modules.

Module 2: Under the Hood with Llama 2 and Python: Understanding How It Works unravels Llama 2's intricacies within Python, guiding you through tokenization, the development of Llama 2 applications via llama.cpp, and parameter adjustments for improved interactions.

Module 3: Building a Llama 2 application begins with a demonstration of zero and few-shot prompting techniques, then moves on to controlling model output for tailored responses. It culminates in practical programming assignments, enabling you to apply your knowledge and showcase your skills in crafting refined Llama 2 applications.

Course materials and assignments will remain open and available for self-paced learning. Learners must earn an overall grade of 80% in order to successfully complete the course and receive the certificate. The passing threshold varies across assignment types. The breakdown below details how grades will be calculated:

Module	Assignment	Percent of overall grade
Module 1	Short Quiz	20%
Module 2	Short Quiz	20%
Module 3	Short Quiz	20%
Module 3	Required Programming Assignment	40%
Module 3	Bonus Programming Activity	0%

Academic Honesty

All submitted work should be your own. Academic dishonesty is not allowed, such as, but not limited to:

- Copying words, ideas, or other materials from another source without giving credit to the original author.
- Copying answers or copying from your peers within the course.
- Employing or allowing another person to alter or revise your work, and then submitting the work as your own.
- Using Artificial Intelligence tools, such as ChatGPT, to create or edit your work and submitting that work as your own, unless you were instructed to use AI as part of the assignment.

Academic dishonesty may result in deleting assignment submissions, forum responses, removal from the course, or other actions as needed. For more information on academic honesty and misconduct, please refer to our [Learner Engagement Policy](#).

Course Support

Questions and discussion of course material should take place within the course itself. Please do not contact instructors or teaching assistants off the platform, as responding to individual questions is virtually impossible.

We encourage you to direct your questions to [Course Issues and Questions](#), where your question might be answered by a fellow learner or one of our course team members. For technical help please contact the [Coursera Learner Help Center](#) support forums.

Accessibility

We are committed to developing accessible learning experiences for the widest possible audience. We recognize that learners with disabilities (including but not limited to visual impairments, hearing impairments, cognitive disabilities, or motor disabilities) might need more specific accessibility-related support to achieve learning goals in this course.

Please use the [accessibility feedback form](#) to let us know about any accessibility challenges such as urgent issues that keep you from making progress in the course (e.g., missing or inadequate alt-text, captioning errors).

Third-party sites and software: While the University of Michigan is not responsible for the accessibility of third-party sites or applications that may be linked from this course, we still encourage you to report associated third-party accessibility issues so that we can ensure you are able to participate. In such cases, we may contact you for additional information as we investigate ways of removing accessibility barriers or to suggest accessible alternatives.

Diversity, Equity, and Inclusion

We welcome all learners to this course. People like you are joining from all over the world and we value this diversity. We strive to create a community of mutual respect and trust, where people from all backgrounds, identities and views are valued and heard without the threat of bias, harassment, intimidation, or discrimination. We pay attention to your feedback, how different types of learners experience this course, and aim to make improvements so the course can best serve everyone. We hope you enjoy learning about topics that are important to you.

Go to next item ✓ Completed

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